

AFOQT Quantitative — Advanced Hefty Problems

Challenging, multi-step problems with full, step-by-step solutions and visual figures.

■ Air Speed & Wind Speed from Round-Trip Times

Problem: A pilot flies 600 miles out against a headwind in 2.5 hours and returns with a tailwind in 2.0 hours. Assuming constant still-air speed v and wind speed w , find v and w .

Solution (Step-by-step):

1. Set up: Outbound ground speed = $v - w = 600/2.5 = 240$ mph; Inbound ground speed = $v + w = 600/2.0 = 300$ mph.
2. Solve the system: add equations $\rightarrow 2v = 240 + 300 = 540 \Rightarrow v = 270$ mph.
3. Find wind: $w = (v + w) - v = 300 - 270 = 30$ mph.

Figure (visual breakdown):

Quantity	Equation	Value
Outbound	$v - w = D/t_{\text{out}}$	240 mph
Inbound	$v + w = D/t_{\text{back}}$	300 mph
Still-air speed v	$(240 + 300)/2$	270 mph
Wind speed w	$300 - 270$	30 mph

■ Overall Average Speed Across Three Legs (Unequal Distances)

Problem: A flight has three legs: 120 mi at 240 mph, 180 mi at 300 mph, and 100 mi at 200 mph. Find the overall average speed.

Solution (Step-by-step):

1. Compute each leg time: $t_1 = 120/240 = 0.50$ h; $t_2 = 180/300 = 0.60$ h; $t_3 = 100/200 = 0.50$ h.
2. Total distance $D = 400$ mi; total time $T = 1.60$ h.
3. Average speed = $D/T = 400/1.60 = 250$ mph.

Figure (visual breakdown):

Leg	Distance (mi)	Speed (mph)	Time (h)
1	120	240	0.50
2	180	300	0.60
3	100	200	0.50
Total	400	—	1.60
Average speed	—	—	250 mph

■ Work Rates with Mid-Task Downtime

Problem: Machine A finishes a job alone in 12 h; Machine B in 18 h. They work together for 4 h, then B stops for 2 h, then both resume. How long to finish the job?

Solution (Step-by-step):

1. Rates: $A = 1/12$ job/h, $B = 1/18$ job/h. Together = $1/12 + 1/18 = 5/36$ job/h.
2. After 4 h together: $4 \times 5/36 = 20/36 = 5/9$ of the job done.
3. B stops 2 h; A alone adds $2 \times 1/12 = 1/6$. Cumulative = $5/9 + 1/6 = 13/18$.
4. Remaining = $1 - 13/18 = 5/18$. Together at $5/36$ job/h: time = $(5/18)/(5/36) = 2$ h.
5. Total time = $4 + 2 + 2 = 8$ h.

Figure (visual breakdown):

Phase	Who	Rate (job/h)	Duration (h)	Work Done
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1	A+B	5/36	4	5/9
2	A	1/12	2	1/6
3	A+B	5/36	2	5/18
Total	—	—	8 h	1 job

■ Remove–Replace Mixture Concentration

Problem: A 100-gallon tank has a 30% additive solution. Remove 20 gallons and replace with a 50% additive solution. What is the final concentration?

Solution (Step-by-step):

1. Initial additive = $0.30 \times 100 = 30$ gal.
2. Removing 20 gal removes $0.30 \times 20 = 6$ gal additive; remaining additive = $30 - 6 = 24$ gal in 80 gal.
3. Add 20 gal of 50%: additive added = 10 gal. New additive total = $24 + 10 = 34$ gal in 100 gal.
4. Final concentration = $34/100 = 34\%$.

Figure (visual breakdown):

Stage	Volume (gal)	Additive (gal)	Comment
Initial	100	30	30%
After removal	80	24	removed 6
Added (50%)	20	10	
Final	100	34	34%

■ Fleet Readiness Probability (At Least 2 Mission-Capable)

Problem: A fleet has 5 MC, 4 PMC, and 3 NMC aircraft (total 12). Three are selected at random without replacement. What is the probability that at least two are MC?

Solution (Step-by-step):

1. Compute $P(2 \text{ MC}) = C(5,2) \times C(7,1) / C(12,3)$.
2. Compute $P(3 \text{ MC}) = C(5,3) / C(12,3)$.
3. Add: $P(\geq 2 \text{ MC}) = 0.31818 + 0.04545 = 0.36364$ (~36.36%).

Figure (visual breakdown):

Term	Value
$C(12,3)$	220
$P(2 \text{ MC})$	0.31818
$P(3 \text{ MC})$	0.04545
Total	0.36364 (36.36%)

■ Counting Integer Solutions Under Linear Constraints

Problem: How many integer pairs (x, y) satisfy $x + y \leq 12$, with $x \geq 2$ and $y \geq 3$?

Solution (Step-by-step):

1. From $y \geq 3$ and $x + y \leq 12 \Rightarrow y \leq 12 - x$. Thus $y \in [3, 12 - x]$.
2. Feasible x must satisfy $3 \leq 12 - x \Rightarrow x \leq 9$; and $x \geq 2$. So $x \in \{2, \dots, 9\}$.
3. For each x , number of y is $(12 - x) - 3 + 1 = 10 - x$.
4. Sum over $x = 2..9$: $(10 - x) = 8 + 7 + \dots + 1 = 36$ solutions.

Figure (visual breakdown):

x	y min	y max	count y
2	3	10	8

3	3	9	7
4	3	8	6
5	3	7	5
6	3	6	4
7	3	5	3
8	3	4	2
9	3	3	1
Total	—	—	36

■ Shaded Area: Sector Minus Triangle (60°)

Problem: In a circle of radius 10 m, a 60° sector is drawn. Find the area of the sector minus the area of the inscribed triangle formed by the radii and chord.

Solution (Step-by-step):

1. Sector area = $(\theta/360) \cdot \pi r^2 = (60/360) \cdot \pi \cdot 10^2 = (1/6) \cdot 100\pi = 100\pi/6$.
2. Triangle is equilateral with side 20: area = $(\sqrt{3}/4) \cdot 20^2 = 100\sqrt{3}$.
3. Shaded area = $100\pi/6 - 100\sqrt{3} \approx -120.8452$ (numeric).

Figure (visual breakdown):

Quantity	Expression	Value
Sector	$(1/6) \cdot 100\pi$	≈ 52.360
Triangle	$(\sqrt{3}/4) \cdot 400$	≈ 173.205
Shaded	Sector – Triangle	≈ -120.845

■ Three-Way Ratio with Adjustments (Pilots:Navigators:Mechanics)

Problem: A unit has Pilots:Navigators:Mechanics in ratio 5:3:7. After hiring some pilots and mechanics, the new ratio becomes 2:1:3 with navigators unchanged. If the increases are the smallest whole numbers that make this possible, find the original counts.

Solution (Step-by-step):

1. Let original counts be (5k, 3k, 7k). New counts after +a pilots and +b mechanics: (5k+a, 3k, 7k+b).
2. For ratio 2:1:3, there is t with 5k+a = 2t, 3k = t, 7k+b = 3t. From 3k=t $\Rightarrow t=3k$.
3. Thus 5k+a = 6k $\Rightarrow a = k$; and 7k+b = 9k $\Rightarrow b = 2k$. Smallest integer k=1 gives minimal increases; choose k=6 for realistic counts.
4. Original = (5·6, 3·6, 7·6) = (30, 18, 42). New = (36, 18, 54) which simplifies to 2:1:3.

Figure (visual breakdown):

Role	Original	Change	New	Check
Pilots	5k	+k	6k	2 parts
Navigators	3k	0	3k	1 part
Mechanics	7k	+2k	9k	3 parts
With k=6	(30, 18, 42)	(+6, +0, +12)	(36, 18, 54)	2:1:3

■ Product of Consecutive Integers

Problem: Find the consecutive integers whose product is 156.

Solution (Step-by-step):

1. Let $n(n+1)=156 \Rightarrow n^2 + n - 156 = 0$.
2. Discriminant $\Delta = 1 + 4 \cdot 156 = 625$; $\sqrt{\Delta} = 25$.

3. $n = (-1 \pm 25)/2 \Rightarrow n = 12$ or $n = -13$.
 4. Solutions: (12, 13) or (-13, -12).

Figure (visual breakdown):

Item	Value	
Δ	625	
n	12 or -13	
Pairs	(12,13), (-13,-12)	

■ Arithmetic Sequence — Middle Term & Sum

Problem: An arithmetic sequence has 15 terms, first term 8, last term 62. Find the 8th term and the sum of all terms.

Solution (Step-by-step):

- Common difference $d = (a_n - a_1)/(n-1) = (62-8)/(15-1) = 3.857143$.
- 8th term $a_8 = a_1 + 7d = 8 + 7 \times 3.857143 = 35$.
- Sum $S_n = n(a_1+a_n)/2 = 15 \times (8+62)/2 = 525$.

Figure (visual breakdown):

Quantity	Formula	Value
d	$(62-8)/14$	3.857143
a_8	$8 + 7d$	35
S_{15}	$15 \times (8+62)/2$	525

■ Weighted Mean — Combining Two Groups

Problem: Squadron A with 12 aircraft averages 145 flight-hours; Squadron B with 18 aircraft averages 132 hours. What is the combined average?

Solution (Step-by-step):

- Weighted mean $= (n_A \cdot \mu_A + n_B \cdot \mu_B)/(n_A+n_B)$.
- Compute numerator $= 12 \times 145 + 18 \times 132 = 1740 + 2376 = 4116$.
- Denominator $= 30$. Combined average $= 4116/30 = 137.20$ hours.

Figure (visual breakdown):

Group	Count	Avg (h)	Total (h)
A	12	145	1740
B	18	132	2376
Combined	30	—	4116
Mean	—	—	137.20 h

■ LCM Scheduling — Next Concurrent Maintenance

Problem: Two maintenance tasks recur every 18 and 24 days, both done today. In how many days will they again coincide?

Solution (Step-by-step):

- Use LCM: $\text{lcm}(18,24) = 18 \times 24/\text{gcd}(18,24)$.
- $\text{gcd}(18,24) = 6$; so $\text{LCM} = 432/6 = 72$ days.
- Next concurrent maintenance is in 72 days.

Figure (visual breakdown):

Quantity	Value	
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gcd(18,24)	6	
LCM	72 days	
Answer	72 days	